For example, if the target temperature was 60°C, the current temperature was 20°C and the tank volume was 100L:

```
Mass of water          = 1kg per Litre
Total Mass             = 100 x 1kg = 100kg
Δ Temperature           = (target temperature - current temperature)
                       = (60°C - 20°C) = 40°C
```

```
Energy required to
heat the water to 60°C = 4182 x 40 x 100 = 16,728,000 J
```

```
Converting Joules in to Wh of heat (divide by 3600):
                       = 16,728,000 J / 3600
                       = 4647 Wh (4.65kWh)
```

If the TankPercent feature is supported, then this estimate SHALL also take into account the percentage of the water in the tank which is already hot.

NOTE The electrical energy required to heat the water depends on the heating system used to heat the water. For example, a direct electric immersion heating element can be close to 100% efficient, so the electrical energy needed to heat the hot water is nearly the same as the EstimatedHeatEnergyRequired. However some forms of heating, such as an air-source heat pump which extracts heat from ambient air, requires much less electrical energy to heat hot water. Heat pumps can be produce 3kWh of heat output for 1kWh of electrical energy input. The conversion between heat energy and electrical energy is outside the scope of this cluster.

9.5.7.5. TankPercentage Attribute

This attribute SHALL indicate an approximate level of hot water stored in the tank, which might help consumers understand the amount of hot water remaining in the tank. The accuracy of this attribute is manufacturer specific.

In most hot water tanks, there is a stratification effect where the hot water layer rests above a cooler layer of water below. For this reason cold water is fed in at the bottom of the tank and the hot water is drawn from the top.

Some water tanks might use multiple temperature probes to estimate the level of the hot water layer. A water heater with multiple temperature probes is likely to implement an algorithm to estimate the hot water tank percentage by taking into account the temperature values of each probe to determine the height of the hot water.

However it might be possible with a single temperature probe to estimate how much hot water is left using a simpler algorithm:

